POINTS TO REMEMBER

- Statistic is a random variable and its probability distribution is called sampling distribution.
- Generally, the random sample used in Statistical Inference is drawn under sampling with replacement from a finite population.
- Standard error is the standard deviation of the sampling distribution.
- Hypothesis is a statement on the population or the values of the parameters.
- Null hypothesis is a hypothesis which is tested for possible rejection.
- Statistical test leads to take decision on the null hypothesis.
- In each statistical hypotheses testing problem, there is one null hypothesis and one alternative hypothesis.
- Type I error is rejecting the true null hypothesis.
- Type II error is not rejecting a false null hypothesis.
- Upper limit of the P (Type I error) is called level of significance, denoted by α.
- Critical region is a subset of the sample space defined by the rejection rule.
- Critical value identifies the elements of critical region.
- If the number of sample observations is greater than or equal to 30, it is called large sample.
- For hypotheses testing based on two samples, if the sizes of both the samples are greater than or equal to 30, they are called large samples.
- For testing population proportion, the sampling distribution of the test statistic is *N(0, 1)*, only when $n \geq 30$, $np > 5$ and $n(1 - p) > 5$.
- For testing equality of two population proportions, the sampling distribution of the test statistic is *N (0, 1)* only when $m \geq 30$, $n \geq 30$, $mp_X > 5$, $m(1 - p_X) > 5$, $np_Y > 5$ and $n(1 - p_Y) > 5$.

EXERCISE 1

7Q1BW9

I. Choose the best answer.

1. Standard error of the sample mean is

   (*a*) σ^2 (*b*) $\frac{\sigma}{n}$

   (*c*) $\frac{\sigma}{\sqrt{n}}$ (*d*) $\frac{\sqrt{n}}{\sigma}$

2. When n is large and σ^2 is unknown, σ^2 is replaced in the test statistic by

   (a) Sample mean (b) Sample variance

   (c) Sample standard deviation (d) Sample proportion

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